DO NOW — RECALL

Lesson 053 · Graphs: Adjacency Lists and Traversal

Name: _____ Class: _____ Date: _____

1. In a BST, where are keys smaller than the parent?

2. What is a graph vertex?

3. How does a queue differ from a stack?

4. An undirected graph has vertices A, B, C, D and edges A–B, A–C, C–D. In vertex order A,B,C,D, write row C of its adjacency matrix and explain why the matrix is symmetric.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

In the left subtree.

2. Model answer

Node representing a point/entity.

3. Model answer

A queue is first in, first out (FIFO); a stack is last in, first out (LIFO).

4. Model answer

Row C is [1,0,0,1], because C connects to A and D. For each undirected edge, both corresponding matrix cells are 1.

By the end, you can…

• An adjacency list records each vertex and its neighbours. For undirected graphs, an edge is represented in both vertices' neighbour lists.

• For BFS, enqueue a starting vertex, mark it visited, and repeatedly dequeue and enqueue undiscovered neighbours.

• Traversal order depends on neighbour ordering. A visited set prevents infinite revisiting when the graph contains cycles.

Vocabulary — know these terms

graph — Vertices linked by edges.

adjacency list — Neighbours indexed by vertex.

BFS — Breadth-first search using queue.

visited — Vertices already discovered.

NEW LEARNING

Read, annotate and explain

Discover

An adjacency list records each vertex and its neighbours. For undirected graphs, an edge is represented in both vertices' neighbour lists.

Trace

For BFS, enqueue a starting vertex, mark it visited, and repeatedly dequeue and enqueue undiscovered neighbours.

Evaluate

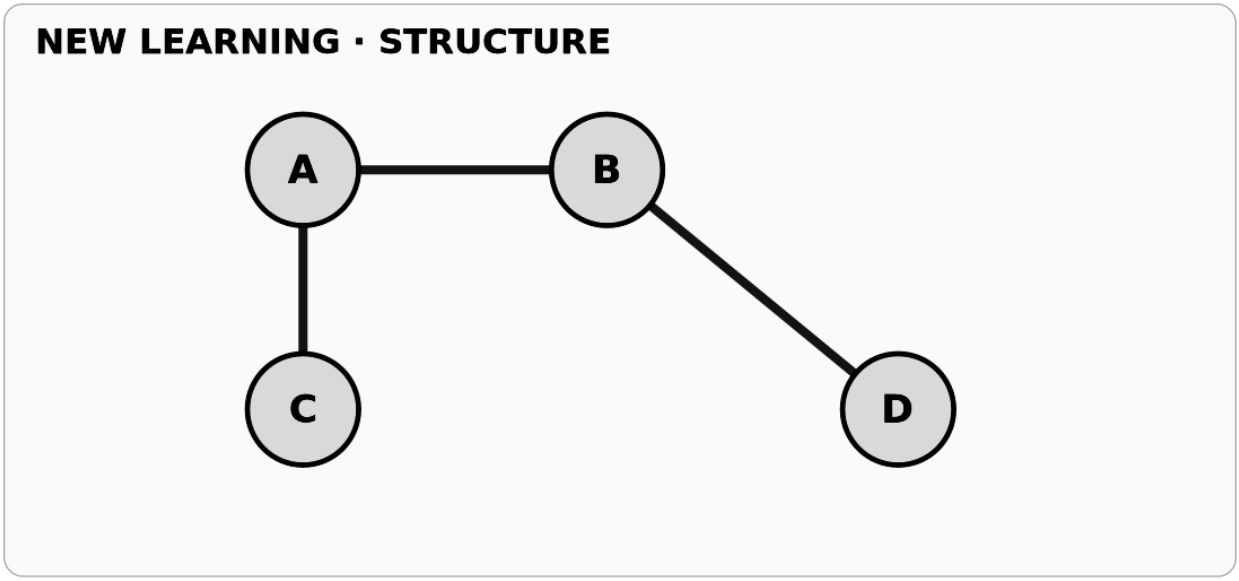
Traversal order depends on neighbour ordering. A visited set prevents infinite revisiting when the graph contains cycles.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



NEW LEARNING · STRUCTURE: A, B, C, D

I DO — WORKED EXAMPLE

Follow the worked method

Trace BFS from A given neighbours A:[B,C], B:[A,D], C:[A], D:[B].

Explain the trace

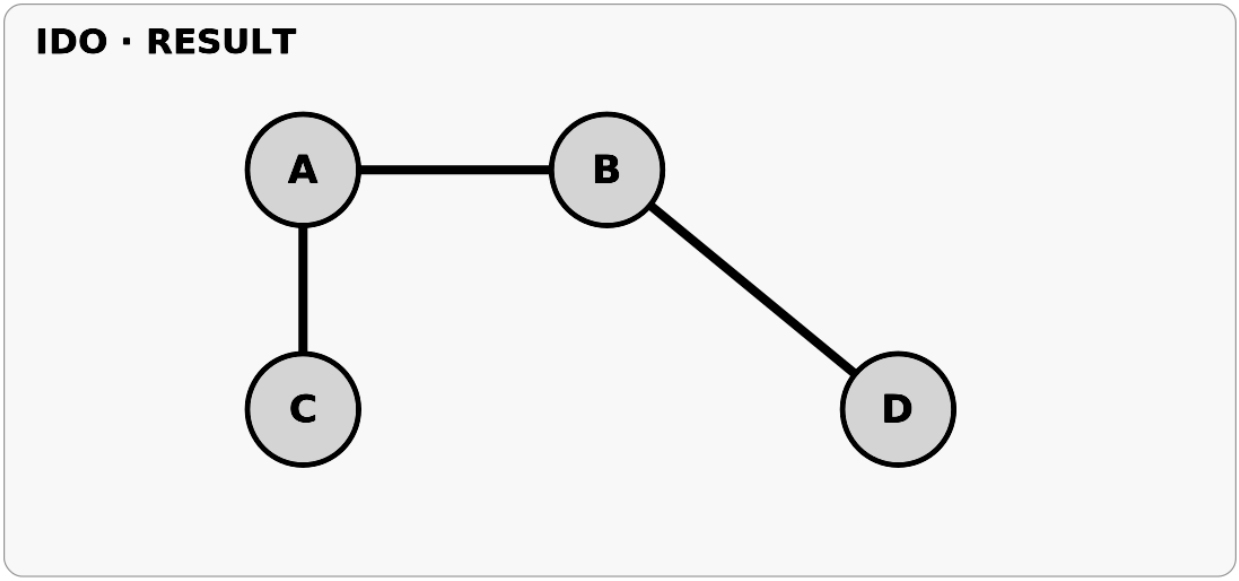
With B before C, BFS visit order A,B,C,D. Each vertex is marked on enqueue.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace BFS from A given neighbours A:[B,C], B:[A,D], C:[A], D:[B].

With B before C, BFS visit order A,B,C,D. Each vertex is marked on enqueue.



IDO · RESULT: A, B, C, D

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Trace a different BFS: X connects Y and Z; Y connects W. Visit from X, Y before Z.
Source code / input:
X:[Y,Z], Y:[X,W], Z:[X], W:[Y]
Queue X → Y,Z → Z,W → W

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Write adjacency lists for undirected edges A-B,A-C,B-D. [6]

Question 2

Trace BFS from A in the lesson graph, recording the queue after expanding A and B. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Explain why DFS can use a stack while BFS uses a queue and why cycles require visited tracking. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

In an undirected graph, A–B is normally stored only under A in a basic adjacency list.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

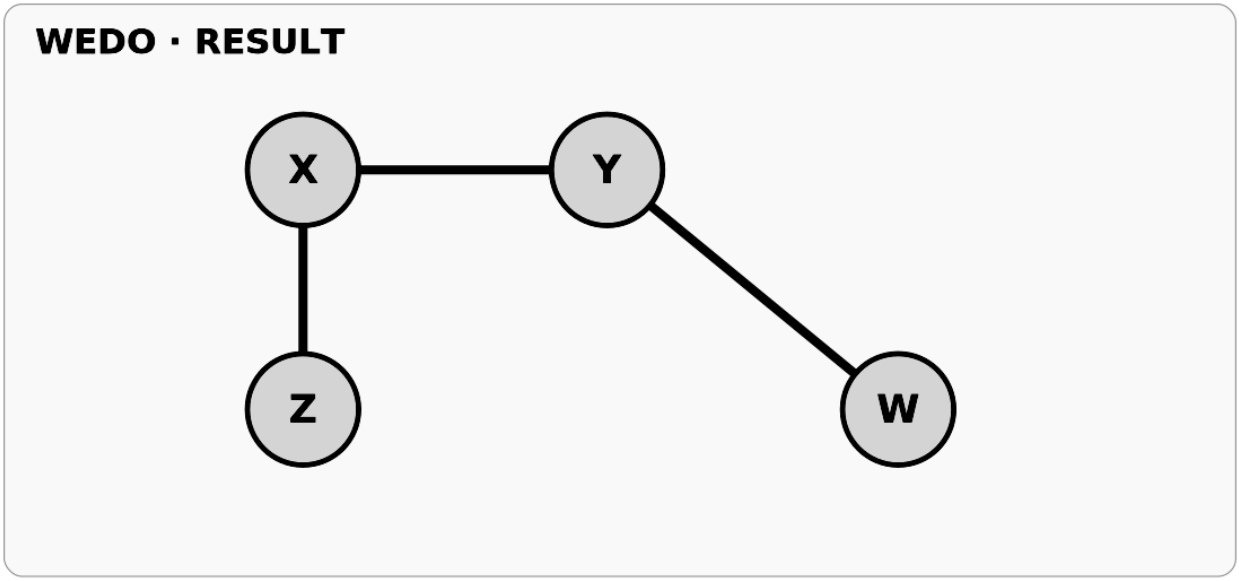
Explain the trace
BFS visit order X,Y,Z,W. FIFO queue processes all distance-one vertices before W.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace a different BFS: X connects Y and Z; Y connects W. Visit from X, Y before Z.

BFS visit order X,Y,Z,W. FIFO queue processes all distance-one vertices before W.



WEDO · RESULT: X, Y, Z, W